

1 Rollout

π_t



GAE

$\hat{A}_r, \hat{A}_{c_k}$

Costs $\bar{J}_k$

2 Rollout graph



nodes = state clusters
edges = transitions
per edge: $\hat{F}, A, C_k$

3 Soft budgets

$$b_k = B_k / L$$

one bound per cost

4 SafeOT projection

new

$$\min_{F \geq 0} -\langle A, F \rangle$$

$$+ \varepsilon \text{KL}(F \parallel \hat{F})$$

$$\text{s.t. } F\mathbf{1} - \gamma F^\top \mathbf{1}$$

$$= (1 - \gamma)\mu_0$$

$$\langle C_k, F \rangle \leq b_k \quad \forall k$$



F^* : target, not executed

5 Actor update

clip c + EMA

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

$$\tilde{A} = \frac{\hat{A}_r - \sum_k \lambda_k \hat{A}_{c_k}}{1 + \sum_k \lambda_k}$$

PPO / TRPO

π_{t+1}

GAE advantages $\hat{A}_r, \hat{A}_{c_k}$

$$\beta_k \leftarrow [\beta_k + \eta(\bar{J}_k - B_k)]_+$$

next rollout